

Adaptive Time Stepping Procedure in Nonlinear Analysis



Presenter Information

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 - **Lead Technical Representative**
(8 years with MSC)
 - Primarily support Marc, Nastran and Patran
 - Also familiar with SimXpert, SimDesigner, Laminate Modeler, MSC.Fatigue ...
 - **Education background**
 - B.S., Mechanical Engineering
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 - Ph.D., Computational Mechanics
- **Joe Satkunanathan @ Santa Ana, CA – Q&A**

Seminar Overview

- Existing time stepping control options for implicit nonlinear analysis
- Unified control option for single and multi-physics jobs: NLSTEP
- New keyword CTRLDEF in NLSTEP – user-friendly entry based on model nonlinearity
- General recommendations
- Demo

Existing stepping control options

- **NLPARM for Nonlinear Static Analysis**
- **TSTEPNL for Nonlinear Transient Analysis**
- **NLPCI for Arc-length Methods**
- **NLADAPT to include additional parameters in NLPARM and TSTEPNL**

NLSTEP for Nonlinear Analysis

- **The NLSTEP entry controls:**
 - Step Size
 - Stiffness Update Strategy
 - Convergence Criteria and Tolerances
 - Frequency of Output
- **It can be used for**
 - Fixed Increment Stepping (FIXED)
 - Adaptive Increment Stepping (ADAPT)
- **It can be used in Static and Transient**
- **It can be used for Mechanical, Thermal and Coupled Analysis**

NLSTEP Bulk Data Entry

NLSTEP	ID	TOTTIME	CTRLDEF						
	"GENERAL"	MAXITER	MINITER	MAXBIS	CREEP		SOL400		
	"FIXED"	NINC	NO						
	"ADAPT"	DTINITF	DTMINF	DTMAXF	NDESIR	SFACT	INTOUT	NSMAX	
		IDAMP	DAMP	CRITTID	IPHYS	LIMTAR	RSMALL	RBIG	
		ADJUST	MSTEP	RB	UTOL				
	"ARCLN"	TYPE	DTINITFA	MINALR	MAXALR	SCALEA	NDESIRA	NSMAXA	
	"HEAT"	CONVH	EPSUH	EPSPH	EPSWH	KMETHODH	KSTEPH		
		MAXQNH	MAXLSH	LSTOLH					
	"MECH"	CONV	EPSU	EPSP	EPSW	KMETHOD	KSTEP	MRCONV	
		MAXQN	MAXLS	LSTOL	FSTRESS				
	"COUP"	HGENPLAS	HGENFRIC						
	"RCHEAT"	SOLVER	DRLXCA	ARLXCA	BALENG	DAMPC	GRVCON	CSGFAC	
		NRLOOP	OUTINV	DTIME1					

NLSTEP	ID		CTRLDEF						
	"LCNT"	NINCC	CONVC	EPSUC	EPSPC	EPSWC	MAXDIVC	MAXBIS	
		MAXITERC	MINITERC				SOL101		

NLSTEP - General

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	GENERAL	MAXITER	MINITER	MAXBIS	CREEP				

- **ID:** invoked by NLSTEP case control command
- **TOTTIME:** total step time to be reached for this step (Default=1.0)
- **MAXITER, MINITER:** maximum and minimum number of iterations in one increment (Defaults = 10, 1)
- **MAXBIS:** maximum number of bisections (Default=10). When negative and reached without convergence the step continues with the next increment (be careful)
- **CREEP:** for creep step (1= creep, 0= no creep)

NLSTEP - MECH

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	MECH	CONV	EPSU	EPSP	EPSW	KMETHOD	KSTEP	MRCONV	
		MAXQN	MAXLS	LSTOL	FSTRESS				

- **CONV:** Flags to select convergence criteria. (Character = “U”, “P”, “W”, “V”, “N”, “A” and combinations; Default = PV).
- **EPSU, EPSP, EPSW:** Error tolerances for displacement (U), load (P) and work (W) criteria. (Defaults = 0.1, 0.1, 0.1).
- **KMETHOD:** Method for controlling stiffness updates. (Character = “PFNT” or “ITER”; Default = “PFNT”).
- **KSTEP:** Number of iterations before the stiffness update for the ITER method. (Default = 10)

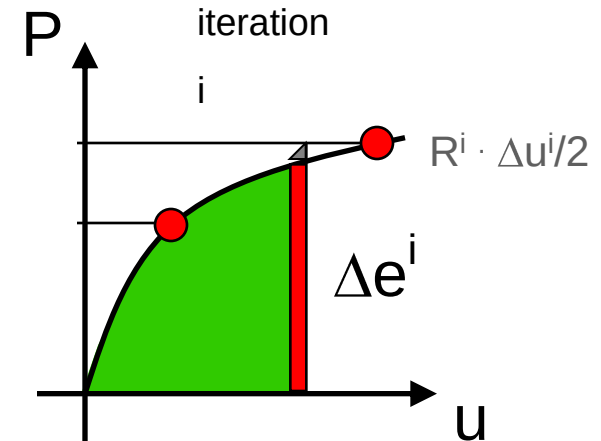
NLSTEP – MECH (cont.)

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	MECH	CONV	EPSU	EPSP	EPSW	KMETHOD	KSTEP	MRCONV	
		MAXQN	MAXLS	LSTOL	FSTRESS				

- **MRCONV: Flag to specify if rotations and moments should be included in the convergence testing when CONV is set to UPV or UPN: (Default = 3)**
 - 0: check on forces, moments, displacements and rotations
 - 1: check on forces, moments and displacements
 - 2: check on forces, displacements and rotations
 - 3: check on forces and displacements

NLSTEP – MECH (cont.)

- Convergence is controlled via **Tolerances** for displacements, residual forces and energy or work



- For displacements and energy these quantities can be relative wrt an increment or absolute wrt the actual solution, i.e. for energy:

PFNT

Δu^i

—

<

EPSU

or

FNT

Δu^i

—

<

EPSU

Δu_{inc}

u_i

NLSTEP – MECH (cont.)

- There are three methods to compute the error:
 - The **weighted average** method – the errors are computed using the weighted average of all dof's of the model. Default.
 - The **vector component (V)** method – convergence checking is performed on the maximum vector component of all dof's in the model.
 - The **length (N)** method – the length of a vector at a grid point is first computed by the SRSS method. Then the error checking is performed on the maximum length of all grid points in the model.

NLSTEP – MECH (cont.)

- Auto Switch (A)
 - if the displacement criterion was requested, but the displacements are getting very small, the program automatically switches to the load criterion.
 - Example: constrained thermal expansion
 - if the load criterion was requested and internal forces are getting very small, the displacement criterion is selected automatically.
 - Example: stress-free motion
 - if displacements and internal forces are getting very small altogether absolute work is chosen.
 - Example: springback

NLSTEP – MECH (cont.)

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	MECH	CONV	EPSU	EPSP	EPSW	KMETHOD	KSTEP	MRCONV	
		MAXQN	MAXLS	LSTOL	FSTRESS				

- **MAXQN:** Maximum number of quasi-Newton vectors. (Default = MAXITER). Not used for PFNT.
- **MAXLS:** Maximum number of line searches allowed for each iteration. (Default = 4). Not used for PFNT.
- **LSTOL:** Line Search tolerance. ($0.01 < \text{Real} < 0.9$; Default = 0.5). Not used for PFNT.
- **FSTRESS:** Fraction of effective stress used to limit the sub increment size in material routines ($0.0 < \text{Real} < 1.0$; Default = 0.2) For non material enhanced elements only.

NLSTEP – ARCLN

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	ARCLN	TYPE	DTINITFA	MINALR	MAXALR	SCALEA	NDESIR	NSMAXA	

- **TYPE:** Constraint type for Arc Length Method. (Character: “CRIS”, “RIKS”, or “MRIKS”; Default = “CRIS”)
- **DTINITFA:** Initial time step defined as a fraction of the load step time (TOTTIME). (Default = .01)
- **MINALR,MAXALR:** Minimum, maximum allowable arc-length adjustment ratio between increments. (Default = 0.25, 4.0)
- **SCALEA:** Scale factor (w) for controlling the loading contribution in the arc-length constraint. (Default = 0.0)
- **NDESIR:** Desired number of iterations for convergence to be used for the adaptive arc-length adjustment. (Default = 4)
- **NSMAXA:** Maximum number of increments. (Default = 1000). The job will stop if this limit is reached.

NLSTEP – FIXED Stepping

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	FIXED	NINC	NO						

- **NINC**, number of increments for fixed stepping. (Integer > 0; Default = 50).
- **NO**, Interval for output. Every NO-th increment will be saved for output. (Integer > 0; Default = 1).

NLSTEP – ADAPTive Stepping

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	ADAPT	DTINITF	DTMINF	DTMAXF	NDSIR	SFACT	INTOUT	NSMAX	
		IDAMP	DAMP	CRITTID	IPHYS	LIMTAR	RSMALL	RBIG	
		ADJUST	MSTEP	RB	UTOL				

- **DTINITF**: Initial time step defined as fraction of total load step time (TOTTIM). (Default 0.01).
- **DTMINF**: Minimum time step defined as fraction of total load step time (TOTTIM). (Default 1e-5).
- **DTMAXF**: Maximum time step defined as fraction of total load step time (TOTTIM). (Default 0.5).
- **NDSIR**: Desired number of iterations per increment. Not used in Dynamics. (Default = 4)

NLSTEP – Adaptive Stepping (cont.)

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	ADAPT	DTINITF	DTMINF	DTMAXF	NDSIR	SFACT	INTOUT	NSMAX	
		IDAMP	DAMP	CRITTID	IPHYS	LIMTAR	RSMALL	RBIG	
		ADJUST	MSTEP	RB	UTOL				

- **SFACT**: Factor for increasing time step. See Remark 3. (Real; Default = 1.2).
- **INTOUT**: Output flag. Integer > -1. (Default = 0)
 - -1: Only the last increment of the step will be output.
 - 0: Every computed increment will be output.
 - > 0: The output will be obtained at INTOUT equally spaced intervals. The time step will be temporarily adjusted if necessary in order to reach these points in time.
- **NSMAX**: Maximum number of increments in the current step. (Default = 99999). The job will stop if this limit is reached.

NLSTEP – Adaptive Stepping (cont.)

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	ADAPT	DTINITF	DTMINF	DTMAXF	NDSIR	SFACT	INTOUT	NSMAX	
		IDAMP	DAMP	CRITTID	IPHYS	LIMTAR	RSMALL	RBIG	
		ADJUST	MSTEP	RB	UTOL				

- **IDAMP**: Flag for activating artificial damping for static analysis. Enter 4 for time step control using damping. Enter 5 for damping based time step control but no damping added. Enter 6 for adding damping when the minimum time step has been reached. Enter 0 for not adding damping. (Integer, Default = 6)
- **DAMP**: Damping ratio. (Real; Default = 2.e-4).

NLSTEP – Adaptive Stepping (cont.)

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	ADAPT	DTINITF	DTMINF	DTMAXF	NDSIR	SFACT	INTOUT	NSMAX	
		IDAMP	DAMP	CRITTID	IPHYS	LIMTAR	RSMALL	RBIG	
		ADJUST	MSTEP	RB	UTOL				

- **CRITTID**: ID of Bulk Data TABSCTL entry which defines the user criteria to use. (Default 0)
- **IPHYS**: Flag to determine if automatic physical criteria should be added and how analysis should proceed if a user criterion is not satisfied. (Default = 2)
- **LIMTAR**: Enter 0 to treat user criteria as limits, 1 to treat user criteria as targets. (Default = 0). Only used if a user criterion is given through CRITTID.

NLSTEP – Adaptive Stepping (cont.)

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	ADAPT	DTINITF	DTMINF	DTMAXF	NDSIR	SFACT	INTOUT	NSMAX	
		IDAMP	DAMP	CRITTID	IPHYS	LIMTAR	RSMALL	RBIG	
		ADJUST	MSTEP	RB	UTOL				

- **RSMALL**: Smallest ratio between increment changes due to user criteria. (Default = 0.1)
- **RBIG**: Largest ratio between increment changes due to user criteria. (Default = 10.0)

NLSTEP – Adaptive Stepping (cont.)

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	ADAPT	DTINITF	DTMINF	DTMAXF	NDSIR	SFACT	INTOUT	NSMAX	
		IDAMP	DAMP	CRITTID	IPHYS	LIMTAR	RSMALL	RBIG	
		ADJUST	MSTEP	RB	UTOL				

- **ADJUST**: Time step skip factor for automatic time step adjustment. Only for dynamics. (Integer; Default = 0).
- **MSTEP**: Number of steps to obtain the dominant period response. Only for dynamics. ($10 < \text{Integer} < 200$; Default = 10).
- **RB**: Define bounds for maintaining the same time step for the stepping function during the adaptive process. ($0.1 < \text{Real} < 1.0$; Default = 0.6).
- **UTOL**: Defines tolerance on displacement used to filter undesirable time step adjustments. ($.0001 < \text{Real} < 1.0$; Default = 1.0)

New Keyword CTRLDEF

1	2	3	4	5	6	7	8	9	10
NLSTEP	ID	TOTTIME	CTRLDEF						
	GENERAL	MAXITER	MINITER	MAXBIS	CREEP				

- Automatically sets up the entries for the time/load stepping adjustment and convergence tolerance
- Four values available as "QLINEAR", "MILDLY", "SEVERELY" , or blank in SOL400
- Three values available as "LCPERF", "LCACCU", or blank in SOL101
- Select a value based on judgment of the nonlinearity of the model

New Keyword CTRLDEF (cont.)

CTRLDEF = QLINEAR

- For linear analysis but with contact nonlinearity
- Smart defaults
 - NINC = 1 (FIXED) or DTINITF = 1.0 (ADAPT) ----- 1 increment load stepping
 - CONV = PV, EPSP = 0.001 and KMETHOD = PFNT
- Good for contact, interference fit and bolt preload

New Keyword CTRLDEF (cont.)

CTRLDEF = MILDLY

- For mildly nonlinear problems
- Smart defaults

NINC = 10 (FIXED) or DTINITF = 0.1 (ADAPT)

CONV = PV, EPSP = 0.01 and KMETHOD = PFNT

- Default load stepping is FIXED
- Good for general nonlinear problem including geometry nonlinearity, material nonlinearity, contact nonlinearity, and their combination

New Keyword CTRLDEF (cont.)

CTRLDEF = SEVERELY

- For severely nonlinear problems
- Smart defaults

NINC = 50 (FIXED) or DTINITF = 0.01 (ADAPT)

CONV = PV, EPSP = 0.01 and KMETHOD = PFNT

Same as the old defaults (blank) except EPSP

- Default load stepping is FIXED
- Good for nonlinear problem involving severe large displacement, large strain, multi-contact pairs

New Keyword CTRLDEF (cont.)

CTRLDEF = LCPERF

- For SOL101 contact problem, prefers performance
- Smart defaults

NINCC = 1

CONVC = PV, EPSPC = 0.001 and KMETHOD = PFNT

- May need to increase the number of increments (NINCC) to achieve optimal results for some contact models

New Keyword CTRLDEF (cont.)

CTRLDEF = LCACCU

- For SOL101 contact problem, prefers accuracy
- Smart defaults

NINCC = 1

CONVC = UPV, EPSPC = 0.01, EPSPC = 0.01 and KMETHOD = PFNT

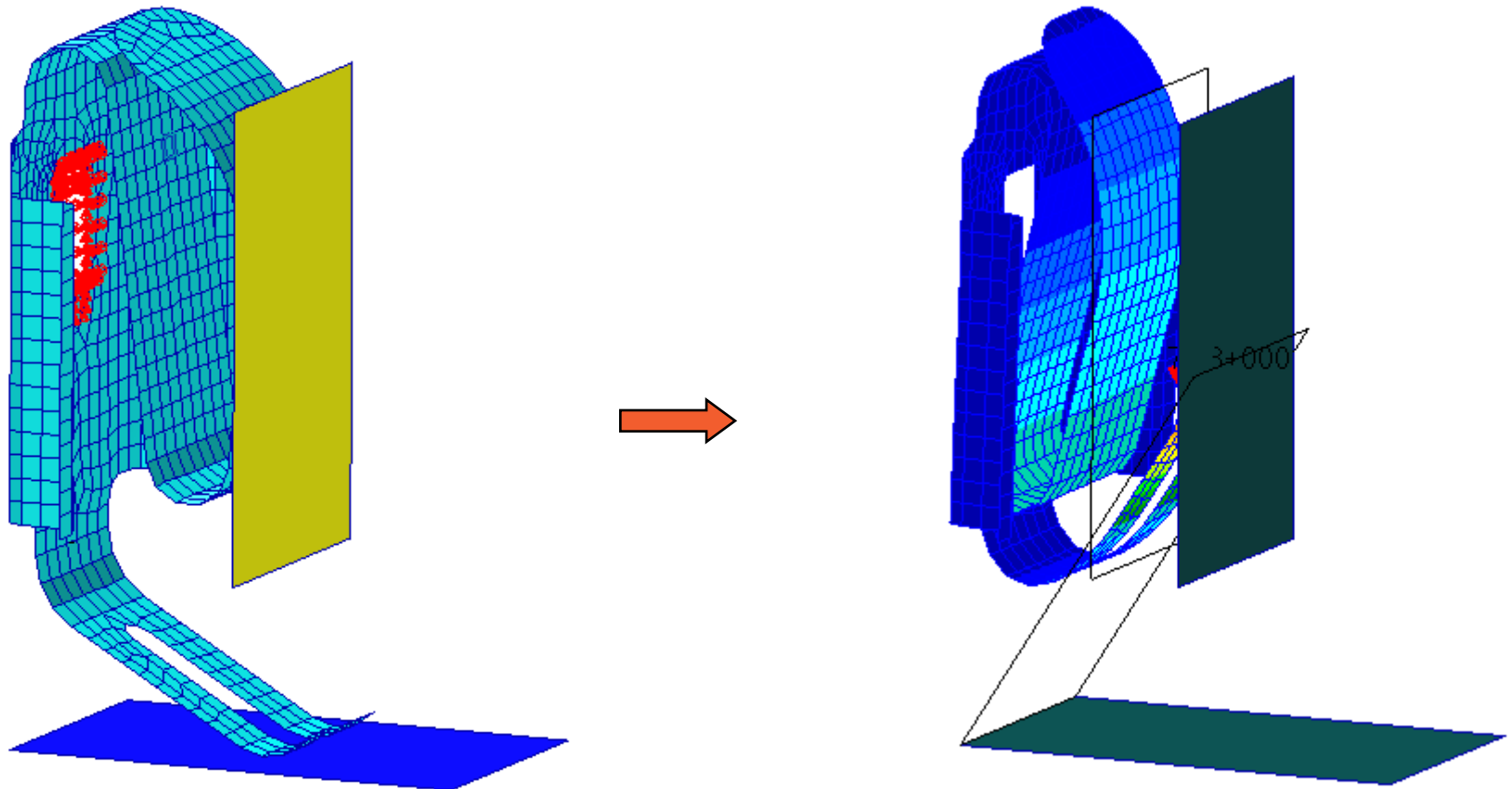
- May need to increase the number of increments (NINCC) to achieve optimal results for some contact models

General Recommendations

- **NLSTEP is a unified load/time stepping control option and should be used for all nonlinear analysis.**
- **Keyword CTRLDEF is a user-friendly entry, users have only to select a value based on their judgment of the nonlinearity of the model to be analyzed.**
- **Adaptive load stepping is usually more efficient.**
- **For the advanced nonlinear elements and materials and contact analysis, FNT or PFNT with CONV=PV should be used.**
- **For contact with friction, PFNT and UPW has been found to be more precise.**

Demo

- Clip model with 3 contact bodies
- Touching contact and large deformation



Demo (cont.)

- **NLPARM with FIXED: 900s**
- **NLSTEP with ADAPT: 163s**
- **NLSTEP with MILDLY: 116s**

Thank You

